

Xuangeng CHU

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🌐 <https://xg-chu.site>

Research Interests

1. Human-centric 3D vision, digital human, human reconstruction and motion generation.
2. Novel view synthesis (NeRF / 3DGS).

Education

The University of Tokyo

Ph.D. Candidate in RCAST, University Fellowship, Supervisor: Prof. Tatsuya HARADA Tokyo, Japan

Princeton University

Visiting Ph.D. Student Researcher, Supervisor: Prof. Jia DENG

Jul 2024 – Sep 2024

Princeton, NJ, USA

Peking University

M.Eng. in Software Engineering, Supervisor: Prof. Yasha WANG

Sep 2018 – Jun 2021

Beijing, China

Tongji University

B.Eng. in Computer Science

Sep 2014 – Jun 2018

Shanghai, China

Work Experience

Tencent, Research engineer

Applied Research Center, work closely with Dr. Ying SHAN

Mar 2021 – Oct 2022

Shenzhen, China / On-site

- *Worked on category extensible object detection algorithm for videos.*
- *Worked on articulated model reconstruction algorithm for a virtual scene generating system.*

Internship Experience

Meta Reality Labs, Research scientist intern

Advised by Shih-En WEI

Aug 2025 – Now

Pittsburgh, PA, USA / On-site

- *Worked on research of audio and style guided motion generation.*

Snap Research, Student researcher

Advised by Dr. Bing ZHOU and Dr. Chuan GUO

Apr 2025 – Now

New York City, NY, USA / Remote

- *Worked on research of motion generation, which was funded by a \$15,000 grant.*

International Digital Economy Academy, Research intern

Advised by Dr. Yu LI

Dec 2022 - Nov 2023

Shenzhen, China / Remote

- *Worked on research and development of human reconstruction and pose estimation.*

Microsoft Research Asia, Research intern

Advised by Dr. Xiulian PENG

Jun 2020 - Feb 2021

Beijing, China / On-site

- *Worked on research of audio-visual speech separation problem (cocktail party problem).*

MEGVII Technology, Research intern

Advised by Dr. Xiangyu ZHANG

Jan 2019 - Jun 2020

Beijing, China / On-site

- *Worked on research and development of object detection in crowded scenes.*

Publication

[NeurIPS 2025] I2-NeRF: Learning Neural Radiance Fields Under Physically-Grounded Media Interactions

Shuhong Liu, Lin Gu, Ziteng Cui, Xuangeng Chu, Tatsuya Harada

- We propose I2-NeRF, a novel neural radiance field framework that enhances isometric and isotropic metric perception under media degradation.

[NeurIPS D&B 2025] Intend to Move: A Dataset for Intention and Scene Aware Human Motion Prediction

Ryo Umagami, Liu Yue, Xuangeng Chu, Ryuto Fukushima, Tetsuya Narita, Yusuke Mukuta, Tomoyuki Takahata, Jianfei Yang, Tatsuya Harada

- We introduce a new dataset for embodied AI, focusing on intention-aware long-term human motion in real-world environments.

[SIGGRAPH Asia 2025] ARTalk: Speech-Driven 3D Head Animation via Autoregressive Model

Xuangeng Chu, Nabarun Goswami, Ziteng Cui, Hanqin Wang, Tatsuya Harada

- We propose an efficient autoregressive framework capable of generating natural 3D facial motions with head poses in real time. Project website: https://xg-chu.github.io/project_artalk.

[CVPR 2025] Luminance-GS: Adapting 3D Gaussian Splatting to Challenging Lighting Conditions with View-Adaptive Curve Adjustment

Ziteng Cui, Xuangeng Chu, Tatsuya Harada

- We introduce a novel approach to achieve high-quality novel view synthesis results under diverse and challenging lighting conditions using 3DGS.

[NeurIPS 2024] Generalizable and Animatable Gaussian Head Avatar

Xuangeng Chu, Tatsuya Harada

- We propose the first generalizable 3DGS head avatar framework that achieves single forward reconstruction and real-time reenactment. Project website: https://xg-chu.site/project_gagavatar/.

[ICLR 2024] GPAvatar: Generalizable and Precise Head Avatar from Image(s)

Xuangeng Chu, Yu Li, Ailing Zeng, Tianyu Yang, Lijian Lin, Yunfei Liu, Tatsuya Harada

- We propose a framework to reconstructs 3D head avatars from one or several images in a single forward pass. Project website: https://xg-chu.github.io/project_gpavatar.

[TVCG 2024] Real-time High-resolution View Synthesis of Complex Scenes with Explicit 3D Visibility Reasoning

Tiansong Zhou, Yebin Liu, Xuangeng Chu, Chengkun Cao, Changyin Zhou, Fei Yu, Yu Li

- We propose a view synthesis method capable of real-time rendering of high-resolution novel-view images from sparse view inputs.

[ICCV 2023] Accurate 3D Face Reconstruction with Facial Component Tokens

Tianke Zhang, Xuangeng Chu, Yunfei Liu, Lijian Lin, Zhendong Yang, Zhengzhuo Xu, Chengkun Cao, Fei Yu, Changyin Zhou, Chun Yuan, Yu Li

- We propose a framework for 3D face reconstruction from monocular images based on 3DMM and transformers.

[CVPR 2020 Oral] Detection in Crowded Scenes: One Proposal, Multiple Predictions

Xuangeng Chu*, Anlin Zheng*, Xiangyu Zhang*, Jian Sun

- We propose a simple and almost cost-free method to improve the detection performance in crowded scenes. Code is available on: <https://github.com/xg-chu/CrowdDet>.

Services

Reviewer, TPAMI; TMM; CVPR; ACMMM; NeurIPS; ICLR; ICML; 3DV.